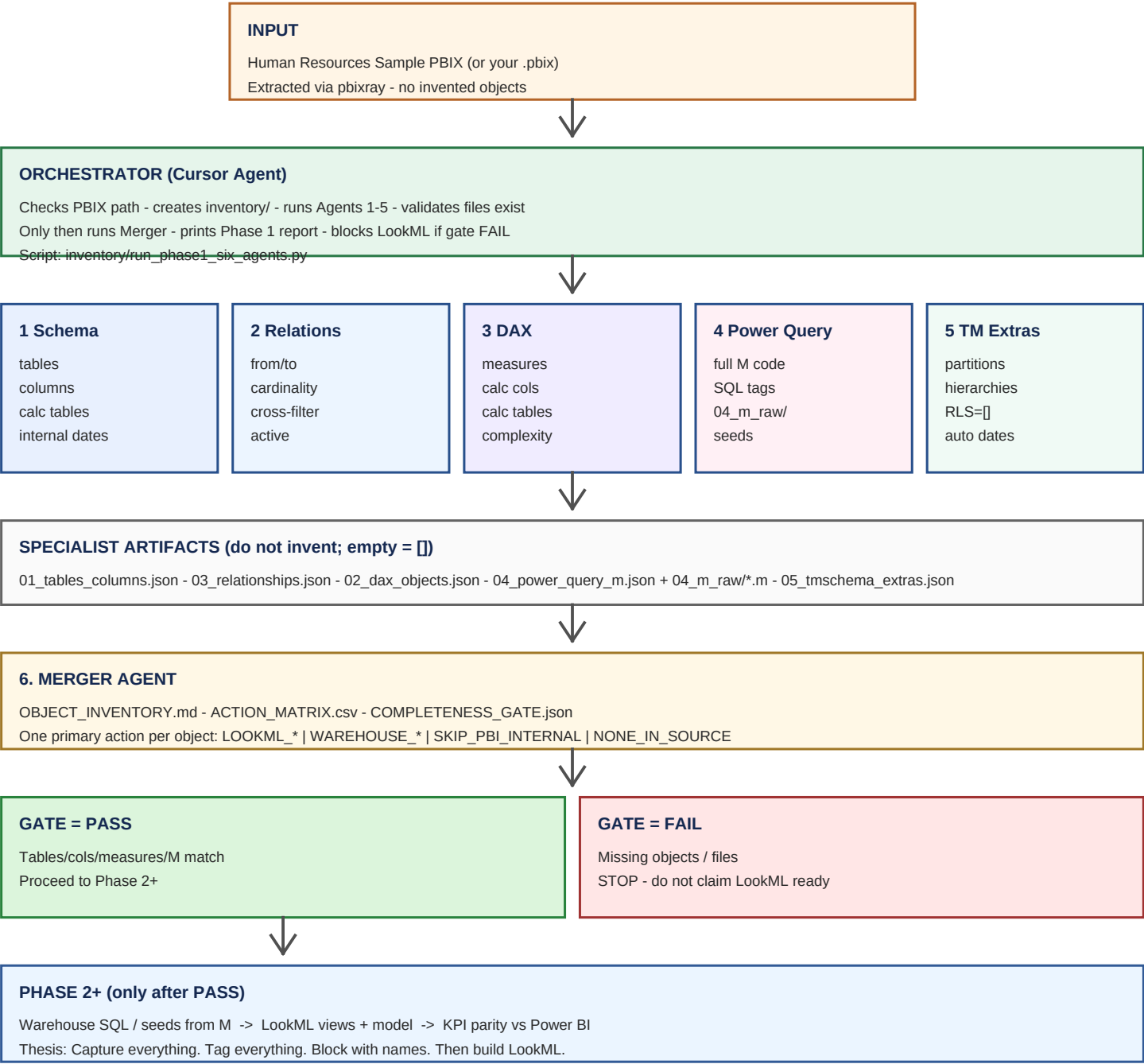


6-Agent Agentic Architecture

Power BI (.pbix) semantic model inventory for Google Looker migration. Reports/visuals/dashboards are out of scope.
Phase 1 = inventory only; LookML and warehouse SQL only after Completeness Gate = PASS.



1. Why an agentic architecture?

A PBIX is not one LookML file waiting to happen. It packs tabular schema, relationships, DAX, Power Query M, and TM extras (partitions, hierarchies, RLS, auto date tables). A single convert-to-LookML pass typically skips M dependencies, stubs complex DAX without disclosure, drops LocalDateTable_* objects, and omits empty categories instead of confirming NONE_IN_SOURCE.

- Specialists = one contract each -> less hallucination / skipped categories
- Empty categories must be listed as empty (e.g. RLS: [])
- Merger enforces a hard completeness gate before LookML is trusted

2. How the system works (runtime)

- Step 1 - Verify PBIX path exists
- Step 2 - Create inventory/ and inventory/04_m_raw/
- Step 3 - Run Agents 1-5 (schema, relationships, DAX, M, TM extras)
- Step 4 - Orchestrator validates all specialist JSON + .m files exist
- Step 5 - Run Merger Agent
- Step 6 - Emit OBJECT_INVENTORY.md, ACTION_MATRIX.csv, COMPLETENESS_GATE.json
- Step 7 - Print Phase 1 report (counts + PASS/FAIL + blockers)

Hard rule

Do NOT generate warehouse SQL or LookML while Completeness Gate = FAIL. Phase 1 purpose is a complete, trustworthy semantic-model inventory.

3. Agent responsibilities

- Agent 1 Schema -> every table/column including LocalDateTable_* and DataTableTemplate_*
- Agent 2 Relationships -> from/to, cardinality, cross-filter, active/inactive
- Agent 3 DAX -> full measure/calc column/calc table expressions + SIMPLE/MODERATE/COMPLEX
- Agent 4 M -> verbatim Power Query to 04_m_raw/<Table>.m + SQL/seed/union tags
- Agent 5 TM Extras -> partitions, hierarchies, RLS/OLS, perspectives, annotations, auto dates
- Agent 6 Merger -> inventory markdown, action matrix, gate JSON

4. Dependency chain the architecture protects

```
M (Power Query) -> warehouse tables / seeds
                  -> calculated columns
                  -> measures / LookML
relationships    -> LookML joins
Date/PeriodNumber -> SPLY, EmpCount, YoY
```

5. HR Sample results (this repo)

- Tables 15 (9 business + 6 internal auto date)
- Columns 87 - Measures 30 - Calc columns 43 - Calc tables 6
- Relationships 8 (all M:1, Single, active)
- Power Query 9 (.m files) - RLS 0 (captured empty)
- Completeness Gate: PASS

6. Known blockers after inventory

- Employee.m must load to warehouse before runnable HR KPIs
- SAMEPERIODLASTYEAR / max PeriodNumber EmpCount / TO % Norm -> LOOKML_TODO_COMPLEX
- KPI parity vs Power BI remains PENDING until data + complex DAX patterns exist

7. How to re-run

```
python3.12 -m venv .venv312
source .venv312/bin/activate
pip install pbixray pandas fpdf2
python inventory/run_phase1_six_agents.py
python generate_architecture_assets.py
```

Full reusable prompt: PROMPT.md. LookML builder guide (after gate PASS): LOOKML_DEVELOPER_GUIDE.md.
Repo: <https://github.com/BasavarajAngadi55/PBI-to-looker-using-cursor->